PulseFit Hybrid Smart Gym and Trainer Booking System

DBMS Level 3 Project Report
Department of Computer Science and Engineering
PES University

Submitted by
Chirag Basanth Bellavi PES1UG24CS133
Chhavi S Wadhwa PES1UG24CS132

Faculty Guide
Dr. Nagasundari S

Abstract

PulseFit is a hybrid gym and trainer booking platform that demonstrates relational, document and vector database patterns in one application. PostgreSQL stores identity, schedules and bookings. MongoDB stores flexible workout logs. PostgreSQL with pgvector ranks trainers and exercises against a member goal. The project prevents trainer double booking with a database exclusion constraint.

Problem Statement and Objectives

Gym systems must protect trainer schedules while recording varied workout details. PulseFit allows members to discover and book trainers, lets trainers publish availability and manage appointments, tracks flexible workouts, and ranks recommendations through vector similarity.

Technology Stack

Layer	Implementation
Frontend	Next.js App Router, TypeScript, responsive CSS and Recharts
Backend	Next.js Route Handlers, Zod and signed HTTP-only sessions
Transactional data	PostgreSQL with pgcrypto, btree_gist and pgvector
Workout data	MongoDB with Mongoose
Similarity	128-dimensional deterministic token and bigram feature hashing

System Architecture

The browser calls protected Next.js route handlers. The application verifies signed sessions and roles before sensitive operations. PostgreSQL owns users, profiles, availability, bookings, audit records and vector metadata. MongoDB owns flexible workout documents and aggregation results.

Relational Schema

Entity	Purpose
users	Authentication identity and role
member_profiles	Member-specific profile data
trainer_profiles	Biography, fee, experience, rating and embedding
trainer_specializations	Normalized trainer skills
trainer_availability	Open appointment time ranges
bookings	Appointment lifecycle
booking_audit	Trigger-maintained change history
fitness_goals	Goal text and optional embedding
exercise_embeddings	Exercise description and vector metadata

Database Integrity

Bookings use a GiST exclusion constraint on trainer ID and a half-open timestamp range. The constraint applies to CONFIRMED bookings and rejects an overlapping appointment for the same trainer, including concurrent requests. The create_booking function locks a matching availability row before creating a booking. A trigger records creation and status transitions in booking_audit. A view summarizes trainer workload.

MongoDB Analytics

Metric	Aggregation approach
Total workouts	Match a member then count documents
Weekly frequency	Group by dateTrunc week
Training volume	Unwind exercises and sets; sum reps times weight
Exercise performance	Group by exercise name
Consistency	Group by workout date then count active days

pgvector Recommendation

Text becomes a 128-value deterministic token and bigram feature-hashing vector. Vectors are normalized and pgvector ranks them by cosine distance. This reproducible local baseline does not require an API key. Its limitation is that it is weaker than a pretrained semantic embedding model.

Security and Authorization

Passwords are verified with bcrypt. Login creates an HTTP-only, SameSite=Lax signed session cookie with 24-hour expiry and Secure in production. Middleware uses Web Crypto for Edge-compatible HMAC verification. API routes enforce roles and ownership before reading or mutating member, trainer, booking or workout data.

Testing Evidence

Executed on 1 November 2026: npx tsc --noEmit and npm test. The seven independent tests passed: valid, expired and tampered session handling, role route policy, deterministic normalized embedding, and wrong vector dimension rejection. PostgreSQL/MongoDB integration tests are pending service availability.

Evaluation Status

Area	Status
TypeScript compilation	Passed
Independent unit tests	Passed 7 of 7
PostgreSQL schema and constraints	Implemented; not service-executed
MongoDB model and pipelines	Implemented; not service-executed
pgvector retrieval	Implemented; not service-executed
Screenshots	Pending live app execution

Area	Status
Production build	Run from unrestricted terminal

Conclusion and Future Work

PulseFit puts transactional scheduling integrity in PostgreSQL, flexible workout tracking in MongoDB and similarity search beside relational metadata with pgvector. Future work can replace the local feature-hashing embedding with a pretrained model, add payments and notifications, and run database integration tests using containers.